

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION
 Question Paper

Roll No. 123104033

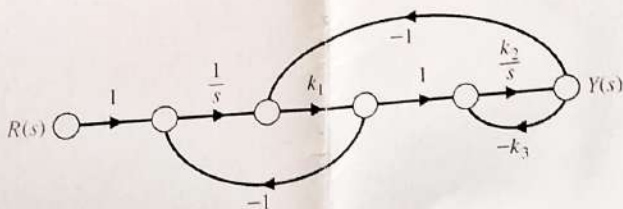
Month and Year of Examination : MAY 2025
 Programme : **B.Tech**
 Course No: **EEPC-208**
 Number of Questions to be Attempted: 5
 Total No. of Questions: 6

Subject: **Control System**
 Maximum Marks: **50**
 Time Allowed: **3 Hrs.**
 Total No. of Pages Used: **2**

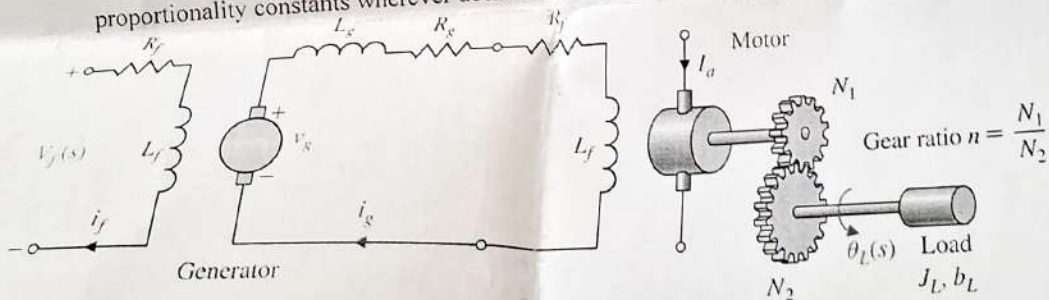
Unless stated otherwise, the Symbols have their usual meanings in context with the Subject. Assume suitably and state, additional data required, if any.
 The Candidates, before starting to write the solutions, should please check the Question Paper for any discrepancy, and also ensure that they have been delivered the question paper of **right course no.** and right subject title.

Note: Question No. 1 is compulsory. Attempt any four questions from Q2 to Q6.

- 1 (a) Discuss two main advantages of closed loop system over open loop system. 2
 (b) Given a system with open loop transfer function $G(s)$ in negative feedback configuration with sensor gain $H(s)$. Obtain expression for the sensitivity of output $Y(s)$ with respect to $H(s)$. 2
 (c) Briefly explain the followings and give example for each 3
 (i) BIBO Stability
 (ii) Relative Stability
 (iii) Asymptotic Stability
 (d) Explain Cauchy's argument principle and its use in Nyquist Stability Criterion. 3
- 2 (a) Obtain the transfer function $\frac{Y(s)}{R(s)}$ of the system with signal flow graph shown as below 5



- (b) Obtain the expression for $G(s) = \frac{\theta_L(s)}{V_f(s)}$ for the following system. Assume necessary proportionality constants wherever deem necessary. 5



- 3 (a) Define and determine various error constants for a unity feedback system with the open loop transfer function as $G(s) = \frac{1}{s(0.5s+1)(0.2s+1)}$ 5
- (b) List various time response metrics for unit step response of a standard second order system and derive mathematical expressions of any three of them. 5
- 4 (a) Evaluate the range of K for which the system $G(s) = \frac{K}{s^5+3s^4+7s^3+14s^2+12s+8}$ is stable under unity feedback. 5
- (b) Sketch the Nyquist plot and determine the stability of the system $G(s)H(s) = \frac{1}{s^2(s+2)}$ 5
5. Draw the Bode plot for a system $G(s) = \frac{400}{s(s+4)(s+10)}$. Define gain margin and phase margin and compute the same for the given system. 10
- 6 (a) For the system with open loop transfer function $G(s)H(s) = \frac{K}{s(s+1)}$; $K > 0$. Design a phase lead compensator using root locus to achieve a settling time of $2\sqrt{2}$ seconds for 2% tolerance band and a damped natural frequency of $\sqrt{2}$ rad/sec. 5
- (b) Discuss the individual effects of proportional, integral and differential controller on the performance parameters of a system. Also discuss what is effect of adding an open loop pole or open loop zero on system stability under unity feedback configuration. 5